7.1 Practice: numbers to a million

← Unit 7.1 · Answers

You don't need to do a whole set at once. Use scrap paper, a place-value table or a number line whenever you like. There is no rush. If a question feels tricky, skip it and come back later.

Set 1 · Warm-up: big numbers

1. What number comes straight after 9 999?
2. Write in digits: twenty-five thousand (*vijfentwintigduizend*).
3. Write in digits: *honderdduizend*.
4. In 64 300, what is the 6 worth?
5. In 482 150, what is the 8 worth?
6. Write in digits: three hundred and six thousand, two hundred (*driehonderdzesduizend tweehonderd*).
7. What is 1 000 more than 249 500?
8. What is 10 000 less than 503 000?
9. Put in order, smallest first: 450 000 · 405 000 · 54 000 · 500 400
10. Write one million in digits.

Talk about it: In question 6, how did you know where to put the zeros?

Set 2 · Warm-up: the number line

1. What number is halfway between 0 and 1 000 000?
2. What number is halfway between 200 000 and 300 000?
3. What number is halfway between 40 000 and 50 000?
4. Count on in jumps of 100 000, starting at 350 000. Write the next three numbers.
5. Count back in jumps of 25 000, starting at 200 000. Write the next three numbers.
6. A number line goes from 0 to 1 000 000 in 10 equal steps. How big is each step?
7. 386 000 is between two "ten-thousands". Which two?
8. Is 386 000 closer to 380 000 or to 390 000?
9. 600 000 + 400 000 =

Talk about it: Draw a number line from 0 to 1 000 000. Where would your town's population go? (Look it up together, or guess.)

Set 3 · Building: rounding (*afronden*)

1. Round 4 300 to the nearest thousand.
2. Round 4 700 to the nearest thousand.
3. Round 4 500 to the nearest thousand.
4. Round 38 250 to the nearest thousand.
5. Round 38 250 to the nearest ten thousand.
6. Round 61 499 to the nearest thousand.
7. Round 254 800 to the nearest hundred thousand.
8. Round 254 800 to the nearest ten thousand.
9. 54 865 people watched a football match. The newspaper rounds to the nearest thousand. What does it print?
10. A new electric bike costs € 1 249. Round the price to the nearest hundred euros.
11. Round 900 to the nearest thousand.

Talk about it: In questions 4 and 5 you rounded the same number twice and got different answers. Are both right?

Set 4 · Building: × and : by 10, 100 and 1000

1. 35 × 10 =
2. 35 × 100 =
3. 35 × 1 000 =
4. 4 800 : 10 =
5. 4 800 : 100 =
6. 72 000 : 1 000 =
7. 60 × 1 000 =
8. 250 × 100 =
9. … × 100 = 9 000
10. 500 000 : 1 000 =
11. A crate holds 100 bottles. How many crates do you need for 4 500 bottles?
12. 7 × 10 =

Talk about it: What happens to the digits when you multiply by 100? Can you explain it without saying “add zeros”?

Set 5 · Stretch: word problems (*verhaaltjessommen*)

1. A town has 86 400 inhabitants. The town next door has 10 000 more. How many people live there?
2. The family car has driven 99 850 km. After a 300 km trip to Groningen and back, what does the odometer show?
3. Ten friends share a lottery prize of € 1 000 000 equally. How much does each friend get?
4. How many times does 1 000 fit into 1 000 000?
5. Use each of the digits 3, 0, 7, 5, 1, 9 once.
   a. Make the biggest number you can.
   b. Make the smallest six-digit number you can (it can’t start with 0).
6. A shop sold 24 000 bags of liquorice (*drop*) in a year. Round that to the nearest ten thousand.
7. On a busy day, 250 trains each carry about 1 000 passengers. About how many passengers is that altogether?
8. True or false: 450 000 rounded to the nearest hundred thousand is 500 000.
9. Which is more: half a million, or 400 000?

Talk about it: In question 7, is the answer exact? Why do people use “about” for numbers like this?

Puzzle corner (optional)

Only if you’re curious. There’s no wrong way to start.

A million seconds. If you counted one number every second, day and night, without stopping, how long would it take to reach 1 000 000? Guess first: a day? a week? a year? Then work it out. (Hint: how many seconds are in one hour? In one day?)

A million sheets of paper. One sheet of paper is about 0,1 mm thick. How tall is a stack of a million sheets? Taller than your house? Than a church tower?

Make your own: invent a “would you rather” question with a million in it. (“Would you rather have a million cents or € 1 000?”)